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data

MDPI

Data Descriptor

Multimodal Dataset of In-Home Physiological and Inertial Measurements from Older Heart Failure Patients

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Abstract

Numerous studies have shown that remote monitoring of heart failure patients can reduce hospital readmission rates and mortality. This dataset includes multimodal physiological and inertial signals (acceleration and angular velocity data) recorded with PerHeart—a remote health monitoring platform intended for heart failure patients. In the pilot, which took place in Poland, 27 participants' health was monitored for one month using the platform with commercially available devices (blood pressure meters, pulse oximeters, bathroom scales, thermometers, and glucometers), resulting in over four thousand physiological measurements. Eight adults were additionally monitored for gait and activity analysis using custom wrist sensors with inertial measurement units, yielding 2536 h of movement data collected over 204 days with almost 690,000 steps detected.

Dataset: https://www.doi.org/10.5281/zenodo.17143199 (accessed on 5 April 2026).

Dataset License: CC BY 4.0

Keywords: remote monitoring; activity detection; heart failure; wrist sensors; inertial signals; gait analysis; ICT platforms

1. Introduction

Heart failure (HF) is a major medical problem associated with high morbidity and mortality among affected patients and incurring a high direct and indirect cost to societies and medical systems around the world [1]. In the European Union, HF accounts for around 2% of current healthcare expenses, estimated at 29 billion EUR annually [2]. The global costs are projected to rise. In the US alone [3], it is projected that the associated costs will reach 70 billion USD by 2030. A significant portion of the costs is due to HF-related hospitalizations.

HF-related hospitalizations can be effectively reduced by remote monitoring of the at-risk patients [4]. Although the performed studies have shown that using even basic telemedicine features such as educational components, self-management, and video communication allows for a reduction in mortality, the full potential of remote monitoring

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can be achieved by employing additional sensors [5]. The sensors can range from constantly worn ECG monitors [6] or smartwatches [7] to incidental use devices such as bathroom scales [8].

Besides physiological parameter monitoring, HF management can be enhanced with additional proxies such as physical activity levels [9] and gait parameters [10]. Both can be assessed based on signals (acceleration, angular velocity) measured using inertial measurement units (IMUs). Although the subject of IMU-based activity and gait detection is very popular and many datasets contain relevant measurement results, datasets targeting the heart failure population are scarce.

In the following paper, we present a multimodal dataset of physiological measurements (blood pressure, blood oxidation, body mass, body temperature, and glucose levels) and inertial signals recorded using a wrist sensor by heart failure patients during their daily activities. The measurements are accompanied by basic user data incl. demographics, self-perceived health status, and life and health satisfaction. The data were gathered during the pilots of the PerHeart (Personalized ICT solution to reduce re-hospitalization rates in heart failure elderly patients suffering from comorbidities) European project [11] undertaken within the Era Permed program [12].

The PerHeart project aimed to develop a modern ICT (Information and Communication Technology) platform for HF patients. The platform's main goal was to reduce re-hospitalization rates by supporting patients with effective disease self-management, giving feedback to caregivers, and providing healthcare professionals with data enabling disease progression monitoring. The platform integrates several applications, medical devices used for physiological measurements (including a digital scale, a blood pressure meter, a digital thermometer, and an oximeter), and a custom wrist sensor recording inertial signals and atmospheric pressure for gait and activity analysis.

In public data repositories, there are several datasets, including remote monitoring physiological measurements or data collected using wrist sensors. A comparison between the described dataset and publicly available ones is presented in Table 1.

Datasets that contain both physiological and inertial measurements are very rare. The only ones known to the authors were described in [13–15]. However, their usability for developing methods in remote health monitoring systems is questionable. The dataset in [14] includes only heart rate and inertial measurements. The data in [13] is simulated. Finally, although the results in [15] were collected with real patient participation, the collected data types are uncommon for digital health platforms.

Remote health monitoring datasets are generally not particularly common, mostly due to the high costs associated with such studies. Notable examples include TIHM [16], RESILIENT [17], and Comprehensive Patient-Health Monitoring Dataset [18]. The data types stored in those datasets differ—each of the studies used a different set of devices, limiting opportunities for cross-study comparisons.

The number of publicly available wrist sensor inertial signals datasets is much higher, as such signals are easier to collect and can serve a plethora of applications like activity recognition [14,19–21], gait analysis [15], or fall detection [14]. In most cases, the datasets are gathered under laboratory conditions [14,15,20] or in a controlled environment like a specially prepared smart home in [19]. During the trials for activity recognition studies, the participants are usually asked to perform a sequence of actions like standing up, walking, turning, and sitting [20] or to move under camera supervision so their actions or positions may be labeled [19]. In gait analysis studies, the participants traverse predefined paths where their gait parameters are recorded using additional equipment [15]. As the duration of lab-based experiments is usually limited, the size of the datasets usually does not exceed a few hours [15,19] of data. Unless specifically targeted, as in [15], where

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Parkinson disease patients' gait was analyzed, participants are usually drawn from readily available sources, such as university students, resulting in a significant age imbalance compared to the general population.

Table 1. Comparison of publicly available datasets.

Dataset	Patients	Age	Physiological Measurements	Motion Mesurements	Monitoring Duration	Location	Targeted Disease
PerHeart (ours)	27	79.3 ± 6.6	heart rate, blood pressure, SpO2, body temperature, body mass, glucose level	acceleration, angular velocity, atmospheric pressure	1 month	in-home	heart failure
Inertial measurement and heart-rate sensor-based dataset [14]	41	27.9 ± 8.6	heart rate	acceleration, angular velocity, magnetic field	one-off session (sequences of activities, falls)	laboratory	fall detection
Dataset for Remote Health Monitoring and Fall Detection [13] [1]	17	60–70	heart rate, SpO2, body temperature	acceleration, rotation	3 h	simulated dataset	fall monitoring
Multimodal Data for the Detection of Freezing of Gait [15]	12	mean 69.1	electroencephalogram, electromyogram, skin conductance	acceleration	one-off session	hospital	Parkinson's Disease (freezing of gait)
TIHM [16]	56 users	70–100	heart rate, blood pressure, body temperature, body mass, body composition, sleep monitoring	in-home movement from domotic sensors	50 days	home	dementia
RESILIENT [17]	73 users	72–99	heart rate, sleep monitoring	step count	25–185 days (varying)	in-home	aging-related comorbidities/ cognitive decline
Comprehensive Patient-Health Monitoring Dataset [18]	n.s. [2]	n.s. [2]	heart rate , blood pressure, respiratory rate , body temperature , SpO2, glucose level	-	4 months	data from monitoring systems	general health
SPHERE [19]	10	18–29	-	acceleration, signal strength (for localization), presence data, anonymized video (location of bounding boxes)	30 min sessions	controlled living environment	none (activity monitoring)
Dataset of Inertial Measurements of Smartphones and Smart-watches [20]	23	44.3 ± 14.3	-	acceleration, angular velocity	one-off session (sequences of activities)	laboratory	none (activity monitoring)
CAPTURE-24 [21]	151	18–53+	-	acceleration	24 h	in-home	none (activity monitoring)

[1] Synthetic dataset. [2] not specified

Datasets collected in real-life contexts are much less common. An example is CAPTURE-24 [21] where the data were gathered by 151 participants who recorded 3883 h of signals with a wrist-worn accelerometer. The samples were labeled thanks to the test subjects wearing cameras.

The following dataset has several features that distinguish it from the other published datasets. To our knowledge, it is the only dataset of physiological and inertial measurements that were recorded during the daily activities of older participants with HF. When it comes to the duration of the user's health parameters monitoring, it was shorter than in the case of other remote monitoring datasets (30 against 25–180 days). The number of participants

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was also limited due to the imposed inclusion criteria (see Section 3.4). These limitations should be considered when using the dataset.

When it comes to the inertial measurements, thanks to significantly longer monitoring compared to other studies (30 days compared to at most 1 day or one-off sessions), the collected data volume is large (2536 h). It enables an analysis of day-to-day changes, which would be impossible in one-time measurements. Contrary to the standard practice of performing only acceleration measurements, the wrist sensors used in this study also recorded angular velocity, enabling more accurate activity detection [22] and atmospheric pressure, enabling the use of more advanced algorithms with additional activity context detection [23].

The data in the presented dataset were collected unsupervised in a non-controlled environment. This ensures that the obtained data are natural and that the users' actions are not constrained by the presence of other people or cameras. At the same time, it prevents the data from being meticulously labeled with activity labels. The signals, however, are accompanied by the step count, enabling the use of the data in supervised gait detection studies.

Although the data is not labeled with the currently performed activities, its volume and collection in natural conditions make it a good candidate for use in weakly supervised and unsupervised learning. After expanding the dataset with some labeled examples, the data may be used in semi-supervised learning. The collected signals can also be jointly analyzed with the step count to label the signals with gait cadence, an indicator of activity intensity [24,25], creating a set of noisy labels that enable the dataset in weakly supervised scenarios. The data can also be used in unsupervised settings, for example, for training autoencoders. Given that the user sample is quite unusual compared to other studies, the collected signals could find application in unsupervised domain adaptation scenarios, where the models need to be adapted to an older population.

2. Data Description

2.1. Dataset Structure

The dataset is available in the Zenodo repository at [26]. The dataset is structured as in the directory tree presented in Figure 1.

```
.
└── Dataset/
    ├── wrist.zip/
    │   ├── user_1/
    │   │   ├── user_1_file_01_inertial_1700980027.parquet
    │   │   ├── user_1_file_01_pressure_1700980027.parquet
    │   │   ├── user_1_file_01_steps_1700980029.parquet
    │   │   └── ...
    │   ├── user_2/
    │   │   ├── user_2_file_01_inertial_1698074792.parquet
    │   │   ├── user_2_file_01_pressure_1698074792.parquet
    │   │   ├── user_2_file_01_steps_1698074794.parquet
    │   │   └── ...
    │   └── ...
    ├── medical.zip/
    │   ├── blood_pressure.csv
    │   ├── body_mass.csv
    │   ├── glucose.csv
    │   ├── oxidation.csv
    │   └── temperature.csv
    ├── personal_questionnaires.csv
    ├── load_dataset.ipynb
    ├── load_dataset.py
    └── perheart_dataset_descriptor.pdf
```

Figure 1. Dataset directory tree. The files with measurements from the wrist and medical sensors are organized into separate directories. The directories were compressed to adhere to the repository's file number limit.

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The dataset contains three types of data files: user demographics information, physiological measurement results obtained with medical devices, and inertial signals captured with a wrist-worn sensor. The user demographics are stored in the personal_questionnaires.csv file. Physiological and inertial signals are stored in medical.zip and wrist.zip archives, respectively.

The data are meticulously described in the perheart_dataset_descriptor.pdf document, which includes a codebook for the results of the personal questionnaires and information on the structure of the measurement files. The repository also contains exemplary code for data loading and exemplary visualization. The code is stored in load_dataset.py and load_dataset.ipynb files. The details are included in Section 4.1.

2.2. Dataset Contents

2.2.1. User Demographics

The users' demographic data are stored in the personal_questionnaires.csv file. The table includes information on the users' sex, age, health state, social and financial situation, and digital literacy. The responses are coded as integers. The dates and optional comments are stored as strings. The codebook for the file is included in perheart_dataset_descriptor.pdf.

2.2.2. Physiological Measurements

The medical measurement results are stored in separate CSV files, one per measurement type. The results are as they were returned by the devices and do not require any prior processing. The list of devices along with the measured values is presented in Table 2. All results are annotated with the user identifier and timestamp in seconds from epoch (UTC). For blood pressure and glucose, the devices returned additional comments, which were encoded as described in perheart_dataset_descriptor.pdf.

Table 2. Physiological data sources in the PerHeart platform. The device models are the ones used in the pilot study for this dataset collection. The detailed information on the demographic data (questions asked to the users and possible answers) can be found in the data dictionary supplied as a part of the dataset.

Data Source	Device Model	Measured Parameters
blood pressure meter	UA-651BLE (A&D Company Ltd, Kitamoto-shi, Japan) or BM 95 (Beurer GmbH, Ulm, Germany)	systolic blood pressure [mmHg], diastolic blood pressure [mmHg], heart rate [bpm]
thermometer	UT-201BLE-A (A&D Company Ltd, Kitamoto-shi, Japan)	body temperature [°C]
blood oximeter	Jumper JPD-500F BLE (Shenzhen Jumper Medical Equipment Co., Ltd., Shenzhen, China)	oxygen saturation [%], perfusion index [-], heart rate [bpm]
bathroom scale	UC-352BLE (A&D Company Ltd, Kitamoto-shi, Japan)	body mass [kg]
glucometer [1]	OneTouch Select Plus Flex (LifeScan Europe GmbH, Zug, Switzerland)	glucose [mmol/L]
demographic data	in-person questionnaires	age, sex, living arrangements, job, self-rated health state, digital skills, HF history

[1] The glucometers were used only by selected users at risk or suffering from diabetes.

The basic statistics of the number of recorded medical measurements and the number of days on which the wrist sensors were used are presented in Table 3.

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https://doi.org/10.3390/data11050106

Table 3. The number of physiological measurements performed by the users during the pilots. Hyphen '-' means that the user did not use the device even once.

User ID	Pilot Days	Wrist Sensor Days	Blood Pressure	Body Temperature	Body Mass	Oxidation	Glucose
1	46	39	90	64	44	64	40
2	29	17	106	-	93	116	7
3	32	18	36	26	22	-	16
4	33	23	41	50	33	66	-
5	37	32	34	37	33	70	-
6	37	30	70	-	50	87	16
7	35	24	18	5	17	2	-
8	28	21	60	60	58	-	-
9	29	-	33	30	39	-	3
10	32	-	76	90	59	1	11
11	31	-	76	67	65	-	-
12	32	-	36	39	47	8	-
13	30	-	34	31	45	-	-
14	11	-	15	9	5	-	2
15	33	-	37	33	29	38	-
16	31	-	34	32	30	29	-
17	28	-	30	43	33	38	-
18	26	-	32	44	36	46	-
19	32	-	35	34	31	30	-
20	28	-	30	30	30	-	25
21	29	-	31	-	35	31	33
22	35	-	43	30	25	32	-
23	30	-	36	34	35	61	-
24	36	-	55	38	35	45	27
25	36	-	38	37	36	52	-
26	34	-	21	22	21	-	-
27	32	-	48	29	32	58	-
mean	31	-	44.3	38	37.7	46	18

Each participant was provided with the working platform for at least a month. For the purpose of the analysis, the pilot duration is expressed as the number of days elapsed between the date the platform was given to the user and the date of the last measurement (the time when the user effectively stopped using the platform).

Regarding platform use, the users generally adhered to the suggested measurement frequency (details in Section 3.4). Apart from users 7 and 26, the participants performed the measurements consistently at least once per day, resulting in finely granulated time series.

All of the users were issued a complete set of devices (a glucose meter was given only to those at risk of diabetes). However, some of the users were selective in their device use. Three users did not measure body temperature, and eight did not use the blood oximeter even once. During the whole pilot study, no significant issues with the platform's accessibility or operation were reported.

2.2.3. Inertial Measurements

The wrist sensor measurements are stored in the Parquet format and are located in the wrist.zip archive. There, they are further separated into user-specific directories. Due to different sensor sampling times, each measurement session is split into three files: inertial measurements, atmospheric pressure, and steps. The files are named using the following convention: user_id_file_file_no_type_ts, where

- id—user identifier (from 1 to 8).
- file_no—file number—two digits with leading 0, e.g., 01, 02 (does not correspond to day of the pilot but to the day when the sensor was in use).
- type—file type, either 'inertial', 'pressure', or 'steps'.
- ts—timestamp in seconds since epoch (UTC, measurements were conducted in Poland, which at the time of the pilots was UTC+1 or UTC+2 for the measurements taken before 29 October 2023).

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Each wrist sensor file type stores an array, with each row corresponding to a single measurement result. The inertial file contains six columns (ax, ay, az for acceleration and gx, gy, gz for angular velocity). The pressure and steps files include a single column with atmospheric pressure and step count, respectively.

The dataset includes files containing a total of 204 full-day captures (complete days when the users wore the wrist sensor). The adherence to performing wrist measurements was generally quite high and ranged between 59 and 86 % of the pilot days (see Table 3). Due to the different adherence of the particular users, the data are not distributed evenly. For example, the number of days when user 1 wore the sensor is more than twice that of user 2.

The inertial signals were collected using a custom wrist sensor developed at Warsaw University of Technology. It includes a BLE-enabled nRF52833 microcontroller, (Nordic Semiconductor ASA, Oslo, Norway) one Bosch Sensortec BMI270 Inertial Measurement Unit [27] (Bosch Sensortec BMI270, Reutlingen, Germany) comprising a 16-bit tri-axial gyroscope and a 16-bit tri-axial accelerometer, and one BMP390 [28] barometer (Bosch Sensortec BMI270, Reutlingen, Germany). It is encased in a Mi Band-strap-compatible 3D-printed case containing a small 220 mAh battery, which allows for over two days of constant operation. The directions of the IMU axes with respect to the sensor's case are presented in Figure 2.

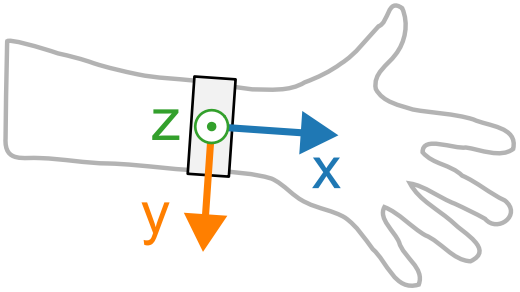


Figure 2. Wrist sensor measurement axes orientation (blue—x axis, orange—y axis, green—z axis). The hand on which the sensor was worn was not enforced.

The device's sensors operated with predefined sampling rates and resolutions, which were constant throughout the whole pilot study and are listed in Table 4.

Table 4. Basic wrist sensor internal sensors parameters.

Sensor	Sampling Rate	Range	Sensitivity
IMU (acceleration)	50 Hz	±4 g	8192 LSB/g
IMU (angular velocity)	50 Hz	±2000 dps	16.384 LSB/dps
IMU (steps)	1.25 Hz	-	1 step
Atmospheric pressure meter	2.5 Hz	-	0.17 Pa [1]

[1] Averaged based on 16 last measurements taken with 200 Hz rate to reduce noise.

For the inertial measurements, the sampling rate was set to 50 Hz, which is a lower value than in typical laboratory-based gait analysis systems [29], but is found in several systems intended for everyday use [30]. Such a decision was influenced by the need to prolong the battery life of the gait sensor and reduce the amount of data needed to be transmitted to the platform's cloud. Our prior studies have shown that such a sampling rate is sufficient to extract basic gait parameters [31].

The gait sensor collects data constantly, even when not being worn. When charging, the sensor periodically dumps data to the tablet so that when the user puts on the device, it stores as few idle measurements as possible. Therefore, each measurement file starts with a short sequence of idle measurement results. If the user decides to take down the device

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during the day and leaves it immobile, it is possible to detect such a situation by analyzing the variability of the measured signals.

In addition to measuring acceleration and angular velocity, the IMU processes this data internally to detect steps taken by the person. The IMU offers two in-house algorithms [27] optimized for wrist-worn applications. The first algorithm is used for step detection and is optimized with detection latency in mind. The second one is used to count the steps. It uses the output of step detection and additionally validates it by considering the step regularity and movement dynamics to prevent false detections. As the piloting study assumed that the activity analysis and the total number of steps were more critical than the timestamps of the steps, the latter option was selected. Figure 3 presents an example of the filtered magnitude of the total acceleration signal and the corresponding output of the step counter.

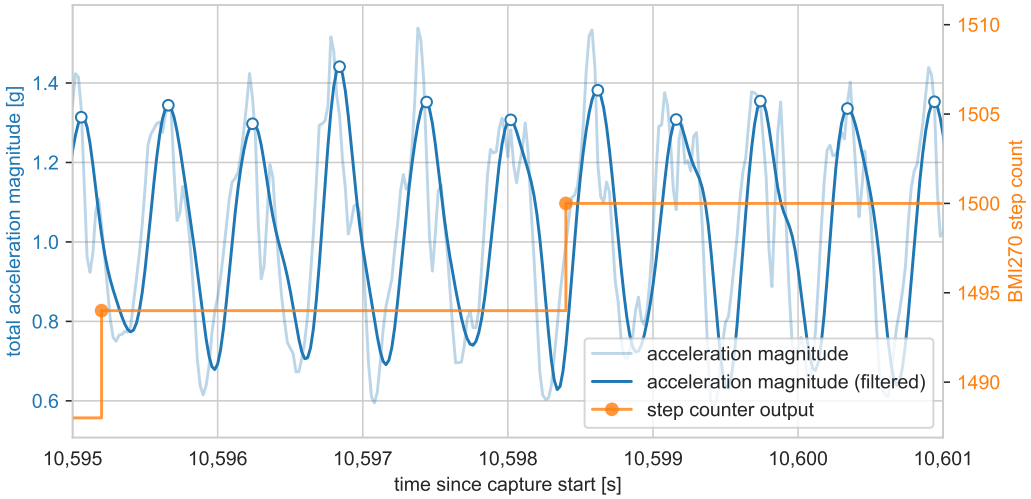


Figure 3. Acceleration magnitude and the corresponding step counter output. The signal was filtered with a low-pass Butterworth filter with a 3.66 Hz cutoff frequency to remove noise and maintain changes related to gait. The peaks detected in the acceleration vector magnitude signal correspond to consecutive steps and can be a basis for step detection [32].

The step counter functionality of the sensor was validated during development, when several team members walked a few hundred meters while counting steps and compared the results with the step counter output. The differences were in the order of 1–2 recorded steps over 300–400 steps made which was deemed as an acceptable error.

The inertial measurements are accompanied by atmospheric pressure measured at a rate of 2.5 Hz. The pressure from BMP390 is returned on demand and is an averaged value from the last 16 measurements taken at a 200 Hz rate, resulting in reduced noise. Thanks to the presence of the barometer, it is possible to detect the change in the person's whereabouts based on the atmospheric pressure changes. If the participant lives on an upper floor, it is possible to detect the moment when the person leaves the flat and comes back based on fast measured pressure changes resulting from the altitude change. It enables adding an additional indoor/outdoor context to the collected data without using energy-inefficient GPS solutions. An example of a person leaving their apartment is presented in Figure 4.

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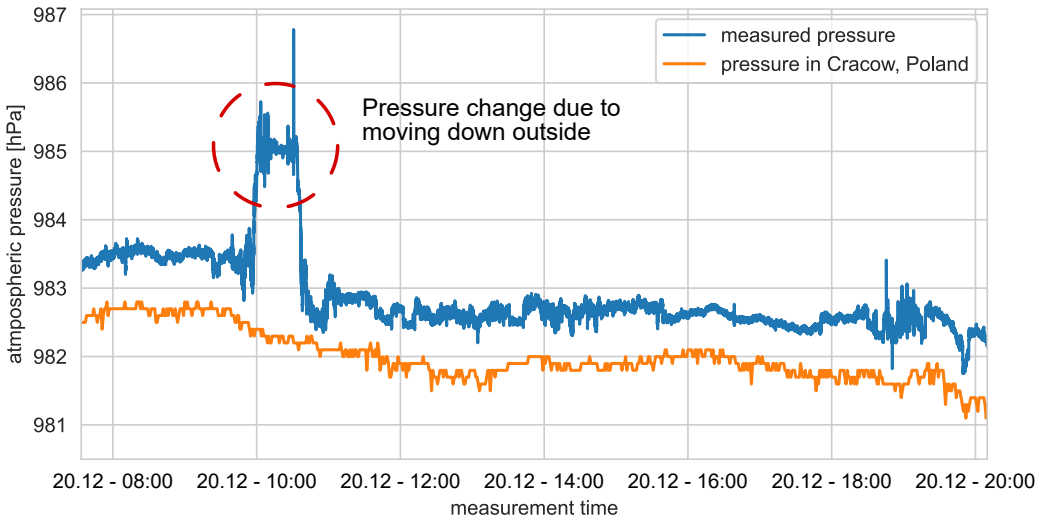


Figure 4. Measured atmospheric pressure changes caused by moving out of the building. The reference pressure curve is based on data from the weather station operated by the AGH University of Krakow [33].

In the presented example, the person living on an upper floor leaves the building, which results in about 1.5 hPa increase in the measured pressure, roughly corresponding to a 12–15 m change in altitude. After about 40 min, the person comes back, which leads to the measured pressure falling.

As the atmospheric pressure changes during the day on its own, moving out should be detected by identifying fast changes rather than comparing current pressure values. The proposed detection method may be harder to apply in the case of people living on lower floors or in houses, as the change in the measured pressure would not be as significant as presented in Figure 4.

3. Methods

3.1. Ethics Statement

This study was accepted by the Jagiellonian University Ethics Committee with reference 1072.6120.17.2023 (date: 15 February 2023). Informed written consent for participation and data sharing was obtained from all participants. All methods in the study were carried out in accordance with relevant guidelines and regulations.

3.2. Participants

The data were collected between July and December 2023, in Poland, with the participation of 27 users (7 female and 20 male) aged between 67 and 94 (mean: 79.3 ± 6.57 years). The participants were patients of the University Hospital in Krakow, Poland. All of them had a history of HF. The basic demographic information for the pilot participants is presented in Table 5.

All participants experienced heart failure events in the two years prior to the pilot's start. User 6 also suffered HF exacerbation during the ongoing pilots. All the tested users were at most stable when it came to rating their health status. Although the sample size in this pilot study was relatively small, the study has provided additional insights into interactions with digital health systems. Thanks to monitoring, it was possible to provide feedback to patients and invite them to in-person visits, leading to successful medication adjustments. The very awareness of the monitoring routine, with the possibility of feedback from the clinician, had a reassuring effect on the patients.

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Table 5. Participants' basic information.

User	Age	Sex	Weight (kg)	Self-Perceived Health [1]	Date of Last Heart Failure Event [2]
1	71	male	80.7	3	04.2023
2	75	male	76.5	3	2021
3	76	female	68.3	3	06.2023
4	81	male	87.5	3	11.2023
5	83	female	52.4	3	2021
6	90	male	64.9	3	2023 [3]
7	69	male	71.8	2	03.2023
8	67	male	75.2	3	06.2023
9	94	female	74.2	2	03.2023
10	74	male	85.2	4	>1 yr. ago
11	83	male	69.1	3	02.2023
12	74	male	98.8	2	06.2023
13	87	male	61.9	2	2023
14	85	male	111.7	2	2022
15	77	male	83.8	3	06.2023
16	83	male	66.3	3	n.s. [4]
17	80	female	63.9	3	2019
18	74	female	91.8	3	summer 2023
19	77	male	83.7	3	2021
20	83	male	104.4	3	2023
21	75	female	67.5	2	02.2023
22	82	male	93.5	3	n.s. [4]
23	77	male	75.1	3	2021
24	78	female	81.8	3	2022
25	85	male	89.5	3	2022
26	72	male	93.8	2	2015
27	88	male	83.8	3	03.2023
mean	79.3	-	79.9	2.8	-

[1] Scored 1–5 (higher is better). [2] The dates are as given by the participants. [3] User 6 also experienced a heart failure event during the pilot trials on 30 December 2023; he did not require hospitalization. The 2023 date in the table corresponds to the event prior to the pilot study. [4] The users did not provide the exact or approximate date of the last HF event.

3.3. Sensors and Devices

The PerHeart platform integrated custom applications, medical devices, and a gait sensor developed at Warsaw University of Technology (WUT). The architecture of the platform setup used in the pilot study is presented in Figure 5.

The main component of the platform is an Android tablet with PerHeart applications installed. The applications gather the results from medical devices and a gait sensor and upload them to the PerHeart cloud for future processing. Users can view their data at any time, as the applications include the required interfaces designed with older adults in mind. All medical devices were wireless-enabled. The list of devices is presented in Table 2.

The medical devices communicate with the tablet using Bluetooth. The tablet application relays the raw measurement results to the PerHeart database without any processing. The wrist sensor used a custom interface connected to the tablet via USB for transmitting the recorded signals as well as charging the device. Photographs of both devices are presented in Figure 6.

The device's sole purpose is to record inertial signals (acceleration and angular velocity) and atmospheric pressure. It is not fitted with any buttons or displays except the diodes signaling low battery level and charging status. The measurement results are saved to the gait sensor's local memory. They are downloaded to the tablet and uploaded to the PerHeart cloud when the user charges the device. The users can monitor their daily step statistics (number of steps, gait cadence) using the supplied application.

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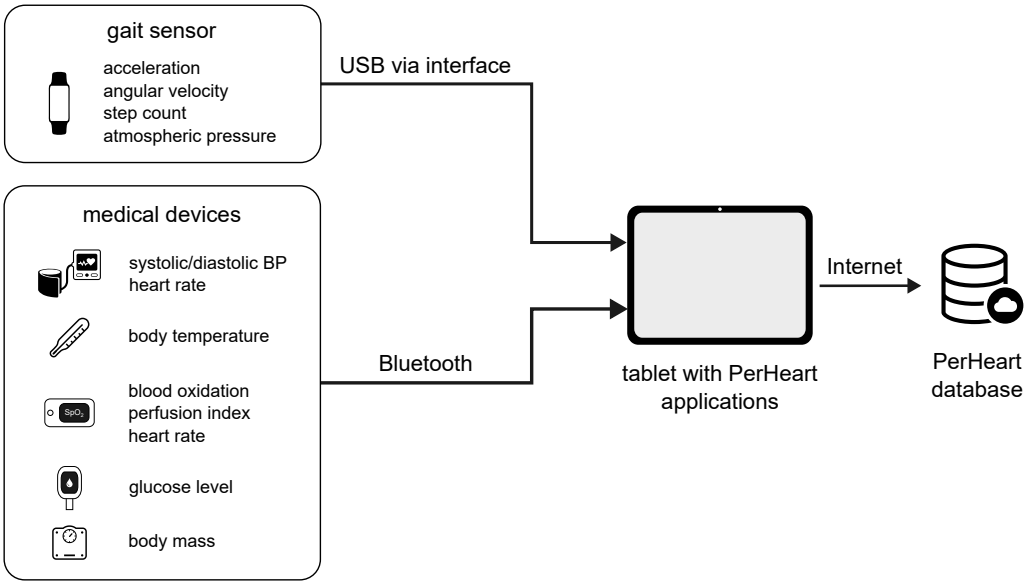


Figure 5. PerHeart platform setup used in the pilot study. The original platform included additional devices and applications, including an intelligent pill box and cognitively stimulating smartphone games. As the dataset does not include any related data, they were omitted from the description.

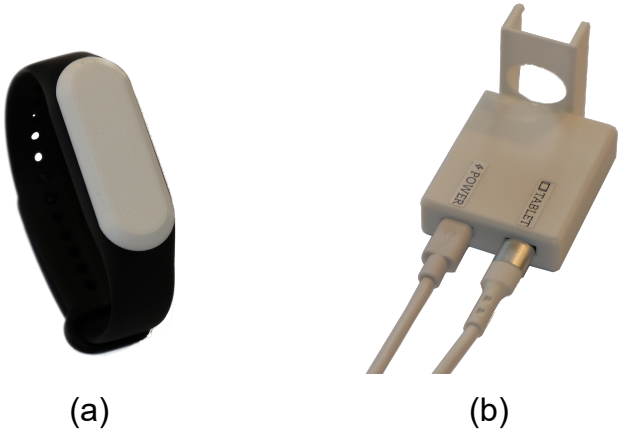


Figure 6. The wrist sensor (**a**) and the charging interface used for data transmission (**b**). Both devices were developed and assembled at WUT.

3.4. Procedure

The patients performed the measurements at their homes in an unsupervised manner. The patients were selected based on the inclusion criteria agreed upon by the PerHeart project partners. The included patients were ⩾65 years of age, had a diagnosis of heart failure based on the available medical records, had had an outpatient visit or hospitalization in the past 30 days, and had agreed to take part. The exclusion criteria included Mini-Mental State Examination < 21/30 or Barthel's index < 91/100.

After signing the consent forms, participants were trained on platform use and provided with the devices. The training included a presentation of the platform capabilities and performing a test measurement session at the doctor's office. Additionally, they were supplied with detailed manuals in Polish covering the use of medical devices and the platform. The users were asked not to alter their daily routine besides performing the measurements and wearing the wrist sensor.

The participants were asked to self-measure the biological signals. Blood pressure was to be measured preferably on the left arm in the sitting position, with the back supported, after five minutes of rest. The measurements were to be taken at least once daily, preferably

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in the morning. However, it was left to the discretion of the caring physician whether the patient was advised to take more measurements during the day, in which case the participant was instructed to follow their physician's advice. Body weight was to be measured at least once per week. The armpit body temperature and the oxygen saturation of blood (peripheral pulsoxymetry method, for at least five seconds) were advised to be taken once daily. The wrist sensor was to be used constantly throughout the day.

Besides encouragement and occasional reminders, the project team did not influence the data collection process. Therefore, the adherence and volume of collected data differed among users.

At the end of the study, the users filled in a questionnaire asking about various issues, including their living arrangements, social and financial situation, self-perceived health, digital literacy, previous heart failure events, and impressions from the project. The participants were not compensated for taking part in the pilots.

3.5. Missing Data Handling

As described in the Section 2.2.3, the wrist sensors used in the study continuously collect data. Therefore, the data in each file are a complete capture between picking up the device and putting it on the charger. Due to the study being conducted in the users' daily living spaces without constant supervision, we had no means of ensuring that the devices were used and that the collected data was uploaded to the platform every day. We have observed situations in which the users did not charge the device, which led to subsequent data loss. Some left it on the charger for longer periods, which resulted in captures including solely idle measurements. Such idle captures were excluded to reduce the dataset size.

In the case of body mass measurements, some corrupted results were observed. They included very low values (a few kg) due to incorrectly performed measurements and occasional measurements performed by the participant's housemates (mass higher by 10 kg than usually observed). Such observations were removed from the dataset to preserve data integrity.

4. User Notes

4.1. Data Preparation

To reduce the dataset size, the timestamps for single wrist sensor measurements are not included, and the measurement results are stored as integers returned by the IMU and barometer. Therefore, data analysis requires prior preprocessing to annotate the samples with timestamps and convert the results to standard units. The timestamp for the given sample can be calculated as

$$t = t_0 + i\frac{1}{f_s}, \tag{1}$$

where t_0 is the timestamp of the first measurement result (in the filename as described in Section 2.2.3), i is the index of the sample in the array, and f_s is the sampling rate (as defined in Table 4). The unit conversion can be performed by dividing the integer value by the sensitivity specified in Table 4.

The repository includes exemplary Python (3.12.4) code for data processing. The codebase includes two files (load_dataset.ipynb notebook and load_dataset.py script). The files include helper functions for data loading, annotation with timestamps, and unit conversion. The notebook includes additional code presenting an exemplary analysis of daily captures for two selected users. Before running the scripts, it is required to unzip the wrist and medical archives.

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4.2. Exemplary Use for Activity Detection

The acceleration measurement results can be used for daily activity analysis. To showcase this possibility, the results were processed using the Biobank accelerometer analysis tool developed at Oxford and described in [34]. The tool can assess the activity intensity and recognize activities using available pre-trained models. As the signals in our dataset are not labeled with specific activities, we analyzed activity intensity and compared it with the step counter's output.

The inertial signals recorded over one selected week for user 2 were preprocessed to meet the analysis tool's unit and timestamp format requirements. Then they were processed using the accProcess tool. The tool analyses the signals in 5 s epochs and outputs an activity volume and intensity summary showing the person's activity during the day. The results of the analysis were compared with gait cadence (the number of steps the person took per minute), which can be treated as a proxy for physical activity intensity [24]. The results are presented in Figure 7.

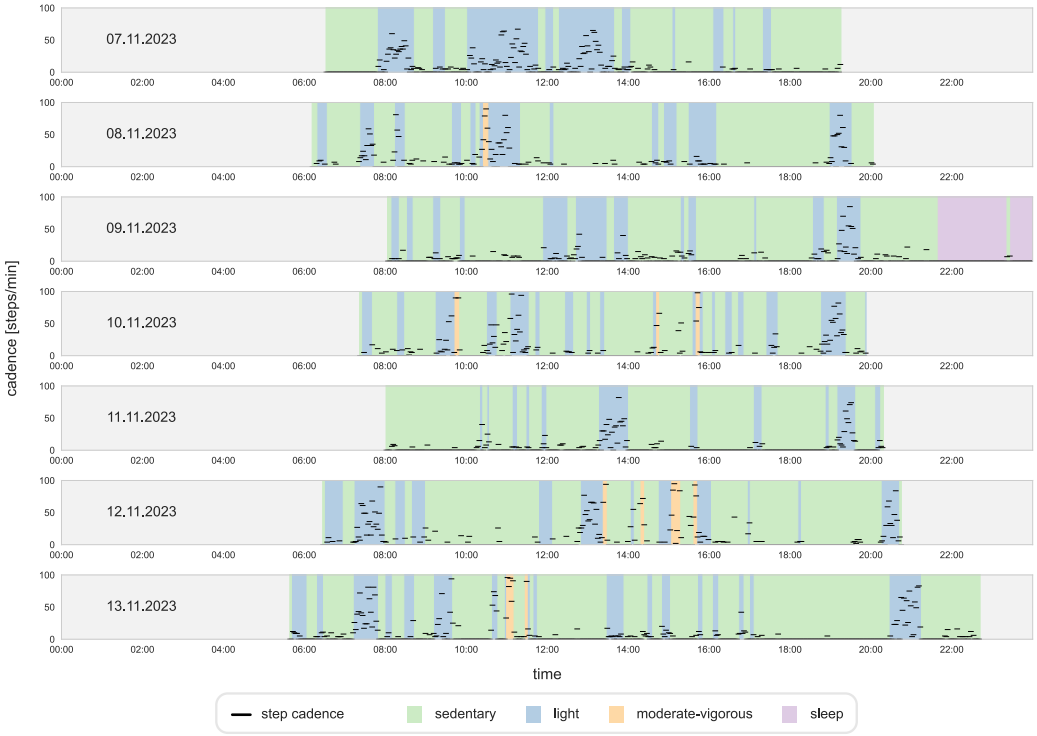


Figure 7. Activity analysis using the Biobank accelerometer tools. The step cadence was obtained based on the step counter output.

During the analyzed week, the users engaged in mostly sedentary and light activities. The step cadence corresponds to the detected activity type—for light and moderate activities, it is visibly higher than for sedentary, which aligns with the findings presented in other works [24,25].

5. Conclusions

In this study, a multimodal dataset of in-home physiological and inertial measurements was presented. The described data was gathered during the pilots of the PerHeart European project with the help of 27 older heart failure patients who received help at the University Hospital in Krakow, Poland. The measurements were performed in an unsupervised manner at the patients' homes for at least one month.

The described collection includes close to 4200 physiological measurements and 2536 h of inertial signals recorded during the users' daily routines (captured by eight users; signals

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during sleep were not recorded). The users generally showed high adherence to the platform use. Although most measurements were performed daily, some users were selective about the devices they used. Several users did not use a body thermometer and blood oximeter even though they were asked to at the start of the study. This shows that when dealing with digital health platforms, such situations should be taken into account. It is especially important when implementing data processing methods and predictive algorithms whose effectiveness may suffer in the wake of missing data.

When it comes to collecting inertial measurements, users did not wear the wrist sensor every day. Given that the sensor was a custom device, which was not actively used and known to the users prior to the pilots, such an outcome was not unexpected. Additionally, for users already wearing the watch, wearing a second bracelet might have caused additional discomfort, leading to lower adherence. Following post-pilot users' suggestions, the newer versions of the device will include a small LCD screen that displays the current time, giving the device an additional watch-like functionality.

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Institutional Review Board Statement: All methods in the study were carried out in accordance with relevant guidelines and regulations. The study was accepted by the Jagiellonian University Ethics Committee with reference 1072.6120.17.2023 (date: 15 February 2023).

Informed Consent Statement: Written informed consent for participation and data sharing was obtained from all participants.

Data Availability Statement: The dataset is available in the Zenodo repository under Creative Commons Attribution 4.0 International license at https://www.doi.org/10.5281/zenodo.17143199. (accessed on 5 April 2026)

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